

Diabetes Related Hospital Readmissions: A Predictive Approach

Author: Alex Nite

Introduction

Hospital readmissions for diabetes are costly and often preventable. Combining patient-level and hospital-level data helps identify high-risk patients and underperforming hospitals. This project analyzes both datasets to uncover patterns, model readmission risks, and guide targeted interventions.

Objective

- Predict 30-day readmission for diabetes patients using patient-level data.
- Identify hospitals features that lead to excess readmission rates.
- Compare model performances to inform clinical and operational strategies.

Data Overview

Both datasets underwent preprocessing which included: handling missing values, basic feature engineering, removal of low-value columns, one-hot encoding, class-weighting for imbalance, normalization, and PCA (90 percent variance retained).

Diabetes Dataset

- 100, K+ patients (85K+ after stratified sampling for KNN specifically).
- Target: binary readmission.

Hospital Dataset

- Includes discharges, readmission rates, ICD and HCPCS codes, and state information.
- Target: excess_readmission_flag.

EDA Highlights

Diabetes Dataset

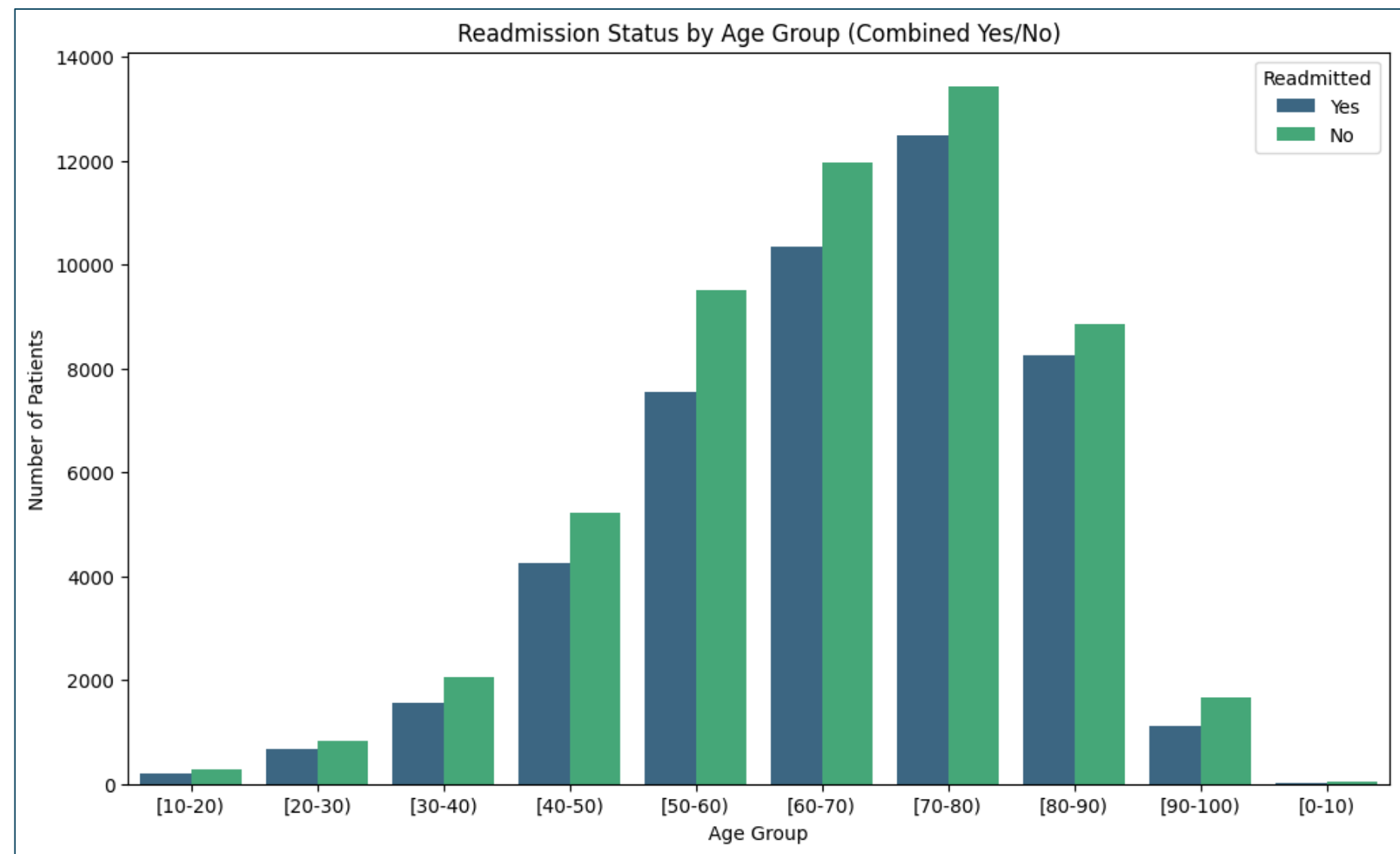


Fig 1.1
Age is a strong predictor of diabetes risk

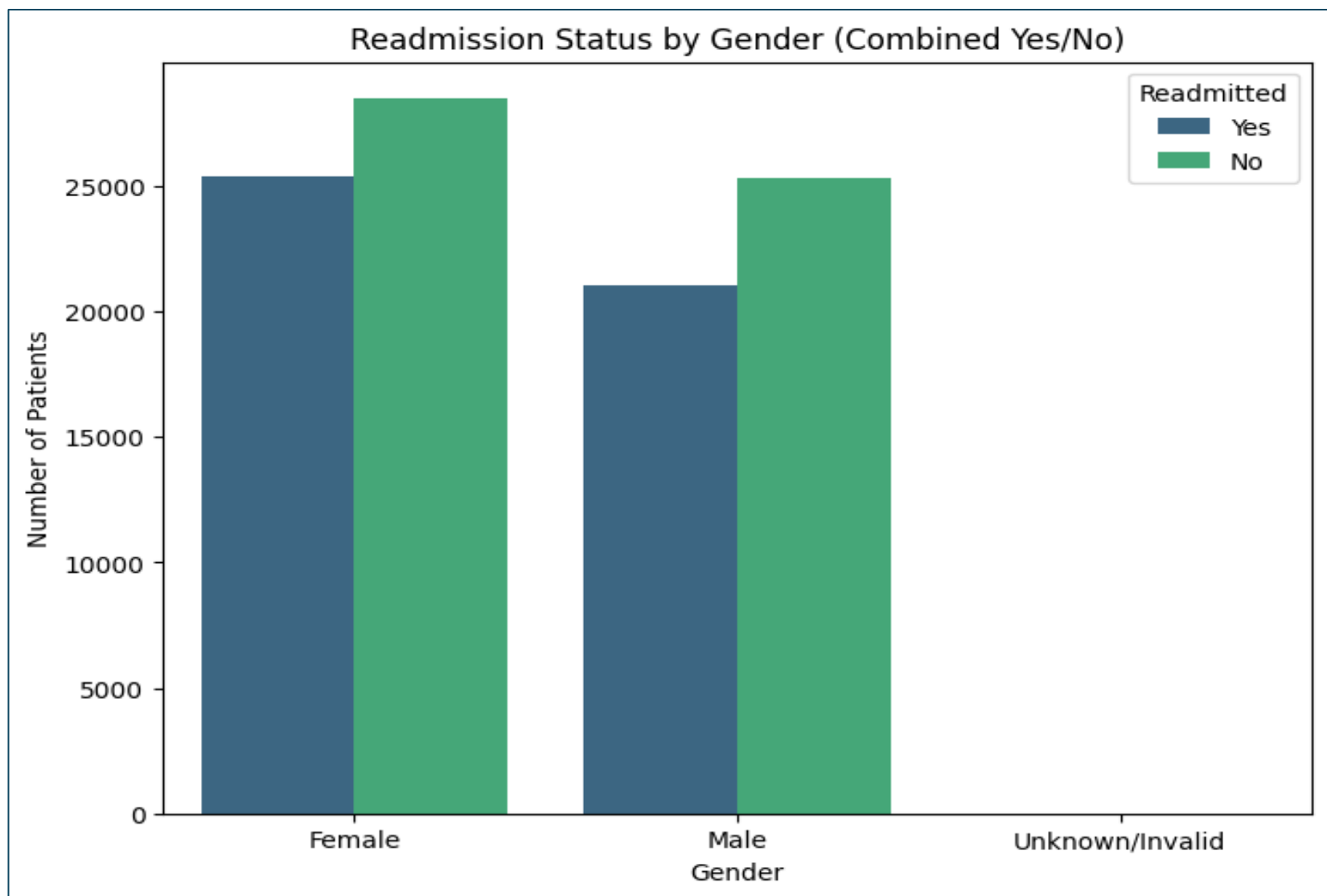


Fig 1.2
Female represent the larger share of the overall patient group.

Hospital Dataset

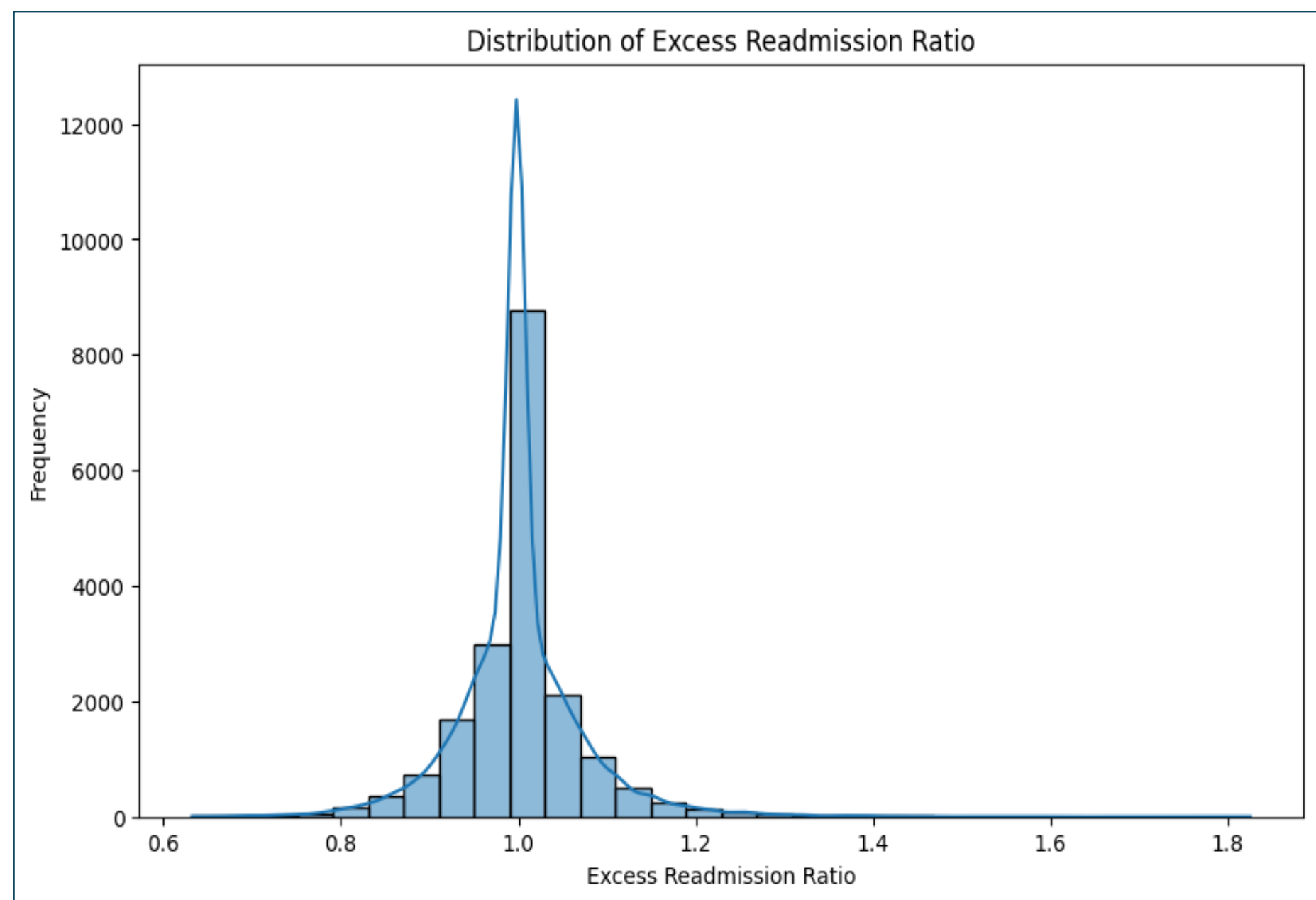


Fig 2.1
Most hospitals cluster around a ratio of 1.0, with a slight right skew showing many just above 1.0 (higher-than-expected readmissions). Fewer hospitals fall below 1.0, indicating lower-than-expected readmissions.

Models

Diabetes Dataset Models

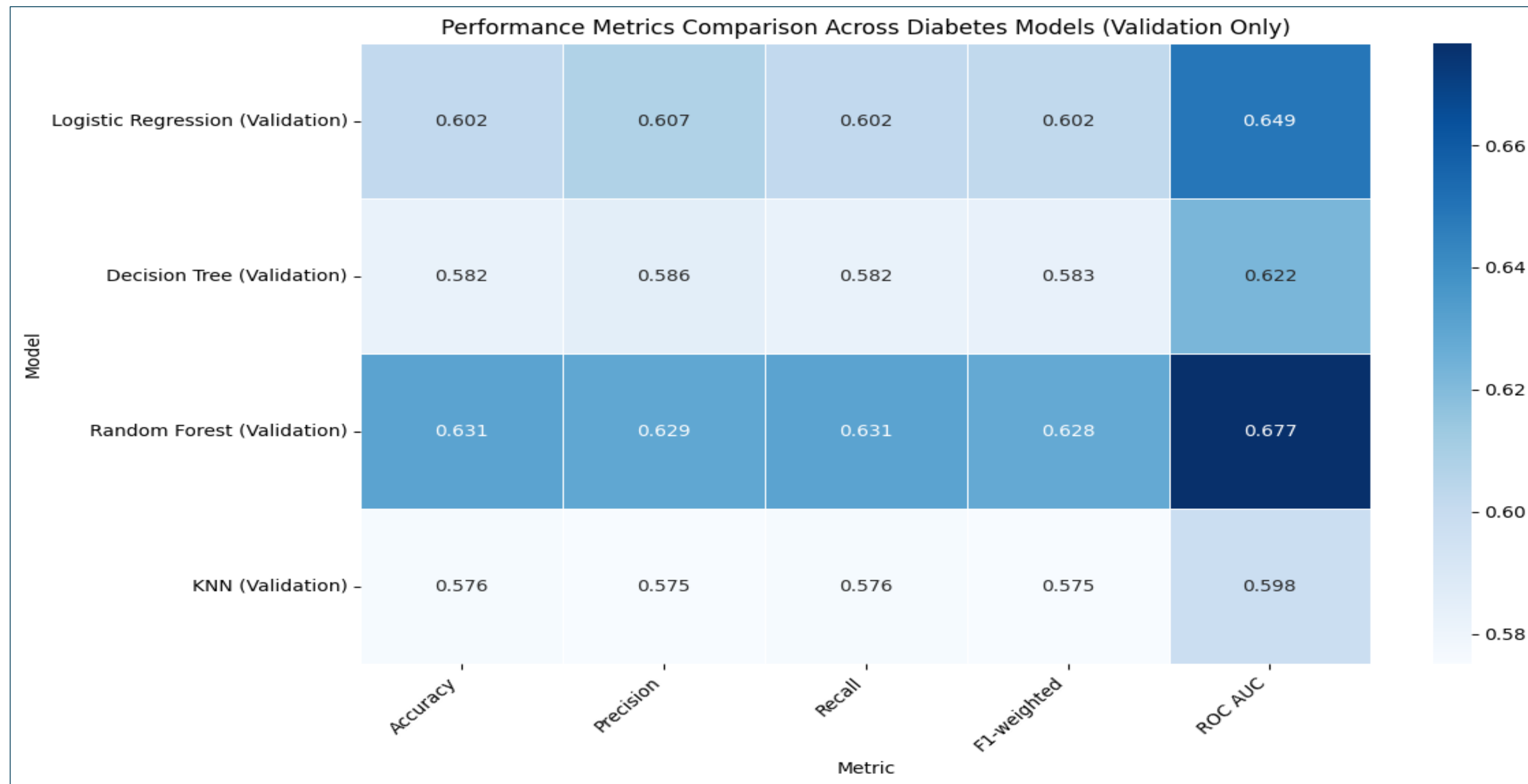


Fig 1.3 Heat Map Comparison of Diabetes Data set Model

Hospital Dataset Model

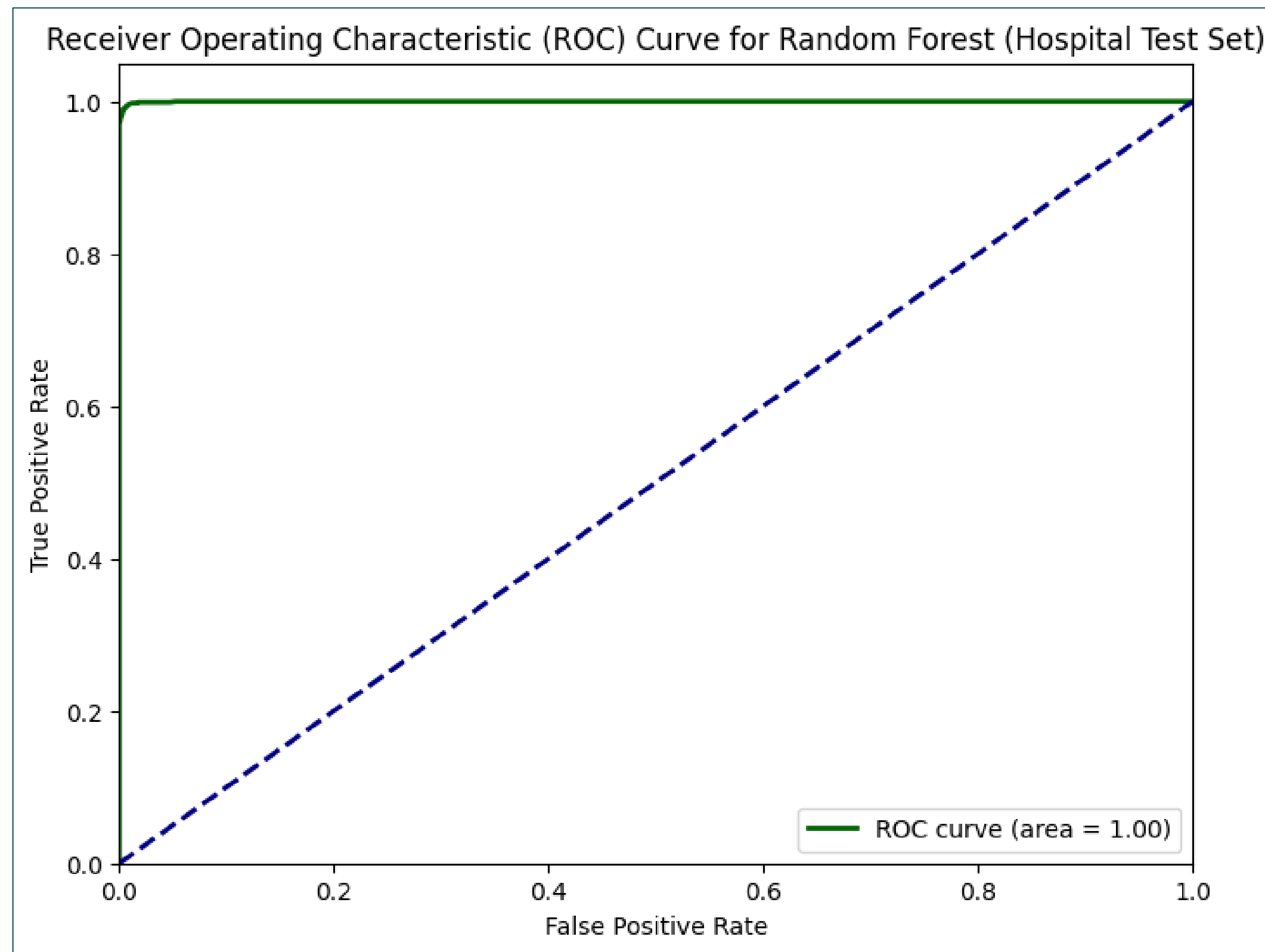


Fig 2.2 ROC curve for Hospital Dataset Test data

Hospital Dataset Random Forest Performance Metric for Test Data

Metric	Precision	Recall	F1-Score
Class 0	0.99	1	0.99
Class 1	0.99	0.99	0.99
Accuracy	-	0.99	-

Table 2.1

Findings & Synthesis

- For patient-level diabetes data, age, prior hospital visits, and medication changes are key drivers of readmission risk. High-risk patients can be identified for targeted care interventions.
- For hospital-level data, the model shows that excess readmissions are largely determined by the hospital's own readmission metrics, shifting the focus from prediction to understanding operational drivers like care coordination, discharge planning, and post-discharge support.
- Overall, combining patient and hospital data provides a holistic view, linking individual risk factors with institutional performance to guide targeted interventions, resource allocation, and policy actions.

Recommendations

- High-Risk Patients:** Target follow-up, home care, and education for patients with complex conditions or multiple prior visits.
- Hospital Processes:** Improve care coordination, discharge planning, and post-discharge support.
- Resource Allocation:** Focus staff and programs where readmissions are most likely.
- Policy Guidance:** Promote standardized protocols and equitable access to follow-up care.